

Using AI-GPU waste heat to keep data-centre electrical rooms dry and clean in the Gulf

Problem, evidence, prior art, market and a demand test

Karthik Nambiar

Sumedh Nitin Jamsandekar

Paul Tiju

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Abstract

Gulf air is hard on data centres. Dust that settles on circuit boards turns conductive once room humidity passes about 60 %, sulfur and chlorine gases corrode the silver and copper on boards, and the hot, humid climate pushes the Middle East's average power usage effectiveness above 1.7 against about 1.5 in North America and Europe. At the same time the newest AI racks (NVIDIA GB200 and GB300, 120 to 142 kW each) are cooled by liquid that returns at about 55 to 65 °C, a steady stream of heat that is mostly thrown away. We propose using part of that heat to regenerate a non-chloride desiccant that, after a cooling coil, keeps the campus electrical rooms (switchgear and UPS) and the air-cooled parts of the racks below the humidity at which dust becomes conductive, with gas-phase filtration, positive pressure and corrosion-coupon control. The principle of drying air with server heat has been proposed since 2014 and tested in laboratories and simulations, but no production deployment was found, every attempt targeted a temperate or mild setting, and the published attempts share documented weaknesses that this design addresses. The UAE grew from 358 MW of data-centre capacity in October 2025 toward about 950 MW forecast for 2028, plus the 1 GW Star-gate UAE cluster inside a 5 GW campus, and a small number of operators decide for hundreds of megawatts. The first test of demand is a set of operator interviews and a free 30-day corrosion and humidity survey of one electrical room. Every figure in this report is taken from the source cited beside it, and our own arithmetic is marked as such.

1 Summary for the deck

- **Problem.** Airborne dust, salt and corrosive gases plus humidity cause short circuits and corrosion in electronics, and dehumidifying hot humid air costs energy. Both are worst in the Gulf [2, 3, 16].
- **Idea.** Heat from the liquid-cooled GPU loop (about 55 to 65 °C) drives a desiccant that keeps the electrical rooms and air-cooled rack parts dry and clean. No electricity is generated, because measured conversion of heat this cool is only 2 to 4 % [28, 29].
- **Why now.** Earlier attempts had heat of 35 to 50 °C and nothing large to dry all year [36, 40, 41]. Current AI racks return hotter water [25, 24], and the UAE is building gigawatt-scale AI campuses in the hottest, most humid, dustiest conditions [47, 48].
- **Who pays.** Data-centre operators. Cooling energy has a price and a regulator in the UAE [17, 18, 19], clean air is a condition of server warranties [7], and a few operators decide for hundreds of megawatts [49].
- **What is new.** No deployed system and no published design was found that uses the liquid-loop heat of AI racks to condition the electrical rooms of a Gulf campus. The parts exist separately (Section 4).
- **Money.** Against the electric desiccant dryers operators already buy, GPU heat saves AED 14,000 to 51,000 a year of electricity per 1,000 m³/h of makeup air, against about AED 560 of pumping. Against an unprotected room, the added filters cost about USD 1,600 to 3,600 a year, the equipment cost of one average outage every 2.7 to 6 years (Section 7).
- **MVP.** Interviews with UAE operators and a free 30-day corrosion-coupon and humidity survey of one electrical room. It measures the problem and tests whether operators will act (Section 10).

2 The problem: what Gulf air does to a data centre

2.1 Corrosion and short circuits

The ASHRAE TC 9.9 committee, written by the IT equipment makers, sets the air-quality limits most data centres design to [1, 2]:

- Gases: copper reactivity below 300 Å per month and silver below 200 Å per month (ANSI/ISA-71.04 severity level G1). Above that, gas-phase filtration of the incoming air is “highly recommended” [2].
- Dust: room air clean to ISO 14644-1 Class 8, with MERV 11 or 13 filters on incoming air [2].
- Humidity: settled dust absorbs water and becomes conductive once the room’s relative humidity passes the dust’s deliquescent humidity, so the guideline says to keep relative humidity below about 60 % [2], and the 2009 paper adds a dew point below 15 °C [1].

Since the change to lead-free solder, the most common open circuit in mission-critical data-centre hardware has been corrosion of the silver terminations in surface-mount resistors, and short circuits come from creep corrosion and ion migration on boards. Air corrosivity comes mostly from sulfur-bearing and partly from chlorine-bearing gases [3]. Equipment makers reported that contamination failures grew as servers spread into polluted markets, especially China with its coal-fired sulfur pollution [4]. A literature review puts realistic worst-case data-centre levels at 10 ppb H₂S, 80 ppb NO₂, 40 ppb SO₂, 2 ppb Cl₂ and 60 ppb O₃ [5]. Across nine Microsoft hyperscale data centres cooled with outside air, relative humidity had the dominant effect on component failures, and disk failures rose at high humidity [6]. The equipment makers Dell, IBM and HP write ISA G1 into their server specifications, and in one telecom case chemical filtration cut corrosion card failures from 36 a month to under 10 [7]. The authors are from a filter maker and IBM, and none of their cases are in the Gulf.

2.2 Switchgear and electrical rooms

High humidity promotes surface partial discharge in switchgear, and manufacturers advise staying below 60 % relative humidity and taking remedies above 75 to 80 % [8]. In the Gulf, salt and desert dust deposits wetted by mist or fog form conductive layers, and in eastern Saudi Arabia insulator contamination is described as the most limiting design parameter for high-voltage insulation [9, 10]. UAE utilities already run sealed control rooms and indoor gas-insulated switchgear with positive-pressure clean-air systems, and desiccant switchgear-room dehumidifiers are sold to DEWA, ADDC and SEC sites, regenerated with electric heaters [11, 12].

2.3 Gulf conditions

- A 2015 dust storm in Doha took PM₁₀ above 7,000 µg m⁻³ [13].
- Our count on hourly 2011 to 2025 weather files for Abu Dhabi International Airport [14]: only 2,317 hours a year (26 %) fall inside the ASHRAE recommended envelope for cooling with outside air (dry bulb 18 to 27 °C, dew point up to 15 °C), none from June to September, and the dew point exceeds 15 °C for 5,171 hours (59 %). Dubai World Central shows 2,500 such hours.
- Drying the outside air would add only about 400 hours (4.5 % of the year) of outside-air cooling, because in summer the air is too hot, not only too humid. A published study likewise found outside-air savings “minimal” in hot dry Dubai [15].

2.4 Energy and water

- The average power usage effectiveness in the Middle East, Africa and Latin America is above 1.7, against about 1.5 in North America and Europe, which the Uptime Institute attributes to hot or humid climates [16]. At 1.7, about 41 % of site power is not IT (our arithmetic).
- DEWA’s marginal industrial rate is 0.38 AED per kWh plus a fuel surcharge of 0.06 [17]. On that basis, 0.1 of PUE on 1 MW of IT load is 876 MWh a year, about AED 385,000 (our arithmetic).
- Abu Dhabi is planning a dedicated cost-reflective tariff for data centres [18]. The federal energy ministry is drafting data-centre efficiency regulations and an energy rating system and has said it could require a technology that cuts data-centre energy by 40 % [19].
- Abu Dhabi’s district cooling uses about 150,000 m³ a day of desalinated water, about 5 % of the emirate’s daily demand [20], and the UAE AI sector could need about 61 billion litres of water a year by 2030 [21].

3 The idea and how it works

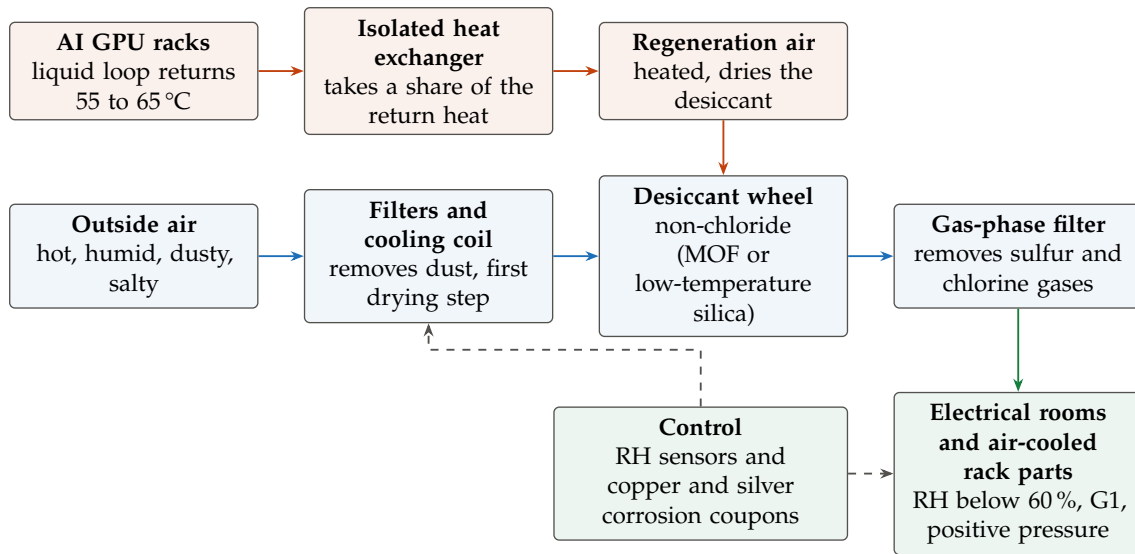


Figure 1: The proposed system. Hot loop (orange), air path (blue), protected space and control (green). The IT cooling loop never depends on the unit: the heat exchanger is isolated and bypassable.

3.1 Why the electrical rooms

Earlier data-centre attempts dried the main server hall. In current AI halls the GPUs and CPUs sit on liquid-cooled cold plates, so much less air passes through the hall and there is little left there to dry, which is one of the documented reasons those attempts did not take hold (Section 4). What remains air-cooled and sensitive all year:

- the electrical rooms. Switchgear humidity risk is sourced above [8]. The same need in UPS rooms is our assumption, to be confirmed in the interviews.
- the air-cooled parts of each AI rack. In the GB200 NVL72 rack the networking, storage and the power supplies (six 5.5 kW supplies per shelf) are air-cooled [22]. A contractor guide estimates about 20 kW per rack still goes to air, a low-authority source [27]. At the LRZ supercomputer centre, the share of node power reaching the water fell from 80 % at 40 °C inlet to under 60 % at 55 °C [40].

3.2 The heat available

- GB200 NVL72: about 120 kW per rack [22], 132 kW per rack with coolant 45 °C in and 65 °C out in an analyst report [24]. GB300 NVL72: up to 142 kW [23].
- NVIDIA's own 2026 description of its next generation: coolant enters at up to 45 °C and leaves at roughly 55 °C [25]. ASHRAE's liquid-cooling classes run to W45 and above, and ASHRAE warns that rising chip power may push facility water down to W32 and W27 [26]. Realistic return water today is about 50 to 55 °C, up to 65 °C in a W45 design.
- Quantity is not the limit. All IT power ends as heat, and at a thermal efficiency of one, 100 kW of heat can remove at most about 147 kg of water an hour (100 kJ/s divided by 2,450 kJ/kg, our arithmetic), which looks large next to the makeup air of an electrical room (our inference, to be checked against real room airflows in the interviews). Temperature, not quantity, is the binding limit.

3.3 The desiccant at these temperatures

- A silica-gel wheel removed 32.6, 37.1 and 40.2 g/h at 50, 60 and 70 °C regeneration, about 19 % less at 50 than at 70 [30].
- A metal-organic-framework (MOF) wheel held 40 to 48 % more dehumidification capacity than silica gel at 50 °C regeneration [31].
- Liquid chloride desiccants (LiCl, CaCl₂) regenerate near 60 °C but risk carryover of salt droplets into the supply air and corrosion. A LiCl unit in Jubail, Saudi Arabia "did not last long in the field" [32], and carryover is described as the gating risk for liquid desiccants [33]. Salt aerosol is exactly what ASHRAE warns against [2], so the design excludes chloride desiccants.

- A solid desiccant wheel removed 0 % of SO₂ at 60 to 79 °C regeneration [34], so a separate gas-phase filter is needed for sulfur and chlorine gases.
- In Gulf summer a desiccant alone cannot dry air that is 45 °C and 24 g/kg to room targets with 55 to 70 °C regeneration. After a cooling coil (air cooled to about 16 °C saturated) the ideal limit is 4 to 6 g/kg, so the desiccant works as a finishing stage that lets the coil run warmer (our psychrometric arithmetic). Desiccant drying also warms the air by about 3.1 °C per g/kg removed [35].

3.4 Against what is sold today

Desiccant dehumidifiers for electrical rooms are already bought in the UAE. Switchgear-room units are sold to DEWA, ADDC and SEC sites [12], desiccant units for data centres hold a 10 to 11 °C dew point [65], and gas-phase filtration is sold to UAE data centres [53]. These dryers regenerate their desiccant with electric, steam or gas heat, typically around 120 °C (the Munters reference reactivation temperature is 121 °C) [35, 66]. A Chinese trade note puts regeneration at over 60 % of such a system’s power, a figure seen only in a search snippet and still to verify [67]. The proposal replaces that heater with GPU loop heat and a sorbent that regenerates at 55 to 65 °C. The buyer keeps a product category it already purchases and stops paying to heat it.

3.5 What the operator gets

- Electrical rooms held below the humidity at which dust conducts, inside ISA G1, measured continuously, which protects equipment and keeps the air inside the class server warranties require [2, 7].
- Less electricity spent on dehumidification by overcooling and reheating, because waste heat does part of the drying. The size of this saving per megawatt is the first number to measure (Section 10).
- Some GPU heat put to use instead of rejected.

4 What exists and where it fell short

Table 1: Prior art on reusing data-centre or electrical heat for dehumidification or sorption cooling. Scale and outcome as reported in each source.

System	Scale	What it did and how it went
Takasago, JP6535445B2, filed 2014 [36]	Patent	Hot-aisle exhaust at 35 to 50 °C regenerates a desiccant rotor that dries outdoor air for offices next to the server room. Design examples only, no product or installation found.
Purdue, 2019 and 2020 [37]	Bench	Silica-gel wheel (about 111 g per cassette) regenerated with 45 °C air from a box warmed by a space heater. Held its target 4.5 h, then supply humidity rose to 63 %. No follow-on found.
Huang et al. 2025 [38]	Lab prototype	50 °C water drives silica-gel coated heat exchangers. Tested four days in a chamber at 15 °C and 90 % RH, which also served as the cold sink. Removed 1.99 to 2.94 g/kg against 7.00 for a heat-pump unit. Payback 3.4 to 5.1 years at 1 RMB per kWh.
Lee et al. 2026 [39]	Lab rig	Liquid desiccant regenerated with heat lifted by a heat pump, one simulated 20 kW rack heated by an electric water heater. Dried outdoor air for the condenser, not the hall. Refrigeration power down 29 % in the hot case, added pumps and fans about 17 %.
LRZ CoolMUC-2, Munich, 2016 to 2024 [40]	Production	Six silica-gel adsorption chillers turned 120 kW of 45 °C heat into about 50 kW of cooling, electrical efficiency about 12. Ran nine years, cooled one storage system. The successor is fully liquid-cooled and drops adsorption.
iDataCool, Regensburg, 2012 [41]	Production	Adsorption chiller idles below 55 °C, about 25 % of the energy reusable as cold water.
Velte-Schäfer et al. 2024 [42]	Calibrated model	At 50 °C a MOF coating raises a chiller from 5.3 to 8.5 kW, output halves from 55 to 45 °C driving heat.
Southeast University, CN201706614U, 2010 [44]	Patent, abandoned	Transformer oil at 65 to 80 °C regenerates a liquid desiccant that dries a substation’s own fresh air.
Pinggao, CN122292093A, 2026 [45]	Patent, under exam	Transformer heat ducted into the switch rooms of a box substation to lower humidity, no desiccant.

Common weaknesses, ranked, and how this design answers each.

1. *Heat too cool.* Adsorption chillers idle below 55 °C and halve from 55 to 45 °C [41, 42]. Answer: use the heat for drying, which works at 50 °C, not for chilling, and take it from current AI loops at 55 to 65 °C.
2. *Hot heat rejection.* Sorption efficiency halves from 24 to 32 °C cooling water [43], and every test used mild conditions. Answer: the cooling coil comes first and the desiccant only finishes the job, so the design does not rely on a cold sink.
3. *No large year-round load.* Earlier outputs cooled leftovers, and full liquid cooling removes the hall load [40]. Answer: serve the electrical rooms and air-cooled rack parts, which need dry air every day.
4. *Low capacity and saturation* [38, 37]. Answer: MOF or low-temperature silica wheels sized as a finishing stage.
5. *Flattering accounting and thin evidence.* Free heat, chillers left out, days of testing [38, 39]. Answer: report total electricity over a full Gulf summer.
6. *Chloride carryover and corrosion* [32, 33]. Answer: solid, non-chloride sorbents only.

5 What is new, and why now

The principle of drying air with server heat is not new (Table 1). What was not found, across papers, patents and vendor material searched in English, Chinese and Japanese:

- a production data centre running server-heat-regenerated desiccant. The nearest commercial step is a hyperscaler pilot of a liquid-desiccant unit on 35 to 60 °C coolant described as likely in Q2 2026, with vendor figures only [46].
- any design using the liquid-loop return of AI racks to condition a data centre's electrical rooms.
- any such work in the Gulf. Both production sites are in temperate Germany [40, 41].

Why now: AI racks of 120 to 142 kW return water hot enough for MOF and low-temperature silica desiccants [22, 23, 24, 31], and the UAE is building gigawatt-scale AI campuses in the conditions where the problem is worst (Section 6).

6 Market

- UAE data-centre capacity 358 MW in October 2025, forecast about 950 MW by 2028. 35 data centres at 98.7% utilisation, a USD 46.1 bn pipeline, 55 % of the GCC's [47].
- Stargate UAE: a 1 GW cluster built by G42, operated by OpenAI and Oracle on NVIDIA GB300, first 200 MW in 2026, inside a 5 GW UAE-US AI Campus [48].
- Khazna: about 300 MW operational, about 370 MW under construction, almost 700 MW live and contracted within 12 to 14 months, operations in-house through NexOps with over 230 staff [49]. Microsoft: 200 MW through Khazna, part of a USD 15.2 bn UAE investment [50]. Moro Hub (DEWA): Warsan green data centre over 100 MW [51]. Khazna's QAJ1 in Ajman, 100 MW, is set to be the first certified liquid-cooled AI data centre in MENA [52].
- A UAE distributor already sells gas-phase filtration and corrosion monitoring to data centres [53], so the category is bought today.
- Scale of spending: Khazna is spending USD 1.3 bn on five facilities adding 210 MW, about USD 6.2 m per MW (our arithmetic) [68]. du is building an AED 2 bn hyperscale site with Microsoft as main tenant [68]. Tabreed, the district-cooling utility, is exploring data-centre investment with Mubadala and Engie [68].
- The market size depends on one unmeasured number: the air-side and humidity load per megawatt that remains in a liquid-cooled AI campus. This is the first thing the MVP asks operators for.

7 Does it save money? A cost model

Two comparisons, each per 1,000 m³/h of outside air supplied to the electrical rooms, because the airflow of a real site is not yet known (it is the first interview question). Weather is hourly Abu Dhabi and Dubai data for 2011 to 2025 [14], and the moisture to remove is our hour-by-hour computation to two targets: a 15 °C dew point (the ASHRAE limit [1]) and a 10 °C dew point (what data-centre desiccant units hold [65]). Our count of hours above a 15 °C dew point in Abu Dhabi (5,171) matches the independent count in Section 2.

7.1 Comparison A: against the electric desiccant dryer operators already buy

Six vendor datasheets give the electricity a desiccant dryer spends on regeneration per kilogram of water removed: Condair DA series 1.21 to 1.58, Munters MCD100 1.28, Munters MX2 80 1.30, Munters DEW-1125 1.76, DST Recusorb 1.39 to 1.50, Bry-Air FFB about 1.43 kWh/kg (our arithmetic on each datasheet's heater power and capacity) [88, 89, 90, 91, 92, 93]. We use 1.30 as typical and 1.21 to 1.76 as the range. Replacing the heater with GPU loop heat removes that electricity.

Table 2: Electricity saved per 1,000 m³/h of makeup air per year. Tariffs: DEWA 0.38 plus 0.06 fuel surcharge [94], Abu Dhabi industrial peak 0.366 from a secondary source, to confirm with ADDC [95].

Site	Target	Water removed (kg/yr)	Electricity saved (kWh/yr, typical)	AED/yr (range)
Abu Dhabi	15 °C dew point	32,286	41,972	15,362 (14,298 to 20,797)
Abu Dhabi	10 °C dew point	55,038	71,549	26,187 (24,374 to 35,453)
Dubai	15 °C dew point	41,346	53,750	23,650 (22,013 to 32,018)
Dubai	10 °C dew point	66,139	85,981	37,832 (35,212 to 51,218)

The added running cost is the pump that moves loop water to the regeneration coil. At the peak hour the regeneration heat is about 30 kW (23.3 kg/h of water at 1.30 kWh/kg), a quarter of one GB200 rack's heat. A 10 K drop needs 0.72 kg/s of water, and at an assumed 100 kPa head and 50 % pump efficiency the pump draws about 145 W, 1,270 kWh a year, about AED 560. The electricity saved is 31 to 92 times the pump's (39,066 to 116,405 against 1,270 kWh). Moisture load peaks June to September, when the loop runs hottest.

Capital: regenerating at 50 °C costs a silica-gel wheel about 19 % of its capacity against 70 °C [30], so the wheel is larger or MOF-based [31]. A brazed plate heat exchanger in the 50 to 600 kW range lists at EUR 720 to 1,780 [96]. At the typical saving, any extra hardware costing less than AED 46,085 (Abu Dhabi, 15 °C) to AED 113,495 (Dubai, 10 °C) per 1,000 m³/h pays back within three years (our arithmetic). This is the ceiling the vendor quote in the MVP is checked against.

7.2 Comparison B: added filters against avoided damage

For a room with no dedicated filtration today, the design adds filters and monitoring. Per 1,000 m³/h per year:

- Gas-phase media: 12.8 kg (an AAF cassette holds 0.5 ft³ of media per 850 m³/h [97]), replaced about annually [98], at USD 11.18 per kg for generic permanganate alumina [99]: about USD 143. Branded media was not priced, so we also carry ten times that, USD 1,434.
- Particle filters: one MERV 13 rigid filter every six months and a MERV 8 prefilter every three months [98], about USD 312 at list prices [100]. One 24 by 24 inch filter per 1,000 m³/h is our assumption.
- Corrosion coupons quarterly: USD 1,128 to 1,813 per room [101].

Total about USD 1,583 to 3,559 a year. Against it:

- In a documented telecom case, chemical filtration cut corrosion failures of switchgear and network cards from an average of 36 a month to about 20, then below 10 [7]. That is at least 26 failures avoided a month at that site.
- Ponemon's 2016 study of 63 data centres put the mean outage at USD 740,357, with USD 9,478 of equipment cost alone per outage [102]. Uptime's 2025 survey found 57 % of impactful outages cost over USD 100,000 and 20 % over USD 1 million, and UPS failures caused 39 % of power-related outages [103].
- Break-even: the yearly filtration cost equals the equipment cost of one average outage every 2.7 to 6.0 years, or the full cost of one average outage every 208 to 468 years (our arithmetic). Avoiding a single board replacement worth about USD 1,600 to 3,600 a year also covers it.

7.3 What the model proves and what it does not

Against the dryers operators already buy, the energy saving alone is AED 14,000 to 51,000 a year per 1,000 m³/h, against about AED 560 of pumping, before any damage benefit. Against an unprotected room, the added filters cost about as much as one board replacement a year, while a documented case avoided at least 26 corrosion failures a month. What it does not prove: the Gulf failure rate without protection, a site's real makeup airflow,

and the share of drying a cooling coil takes before the desiccant. The savings scale linearly with all three, and the MVP survey and interviews measure them.

8 Why operators would take the initiative

1. **Cooling energy has a price and a regulator.** PUE above 1.7 in the region [16], about 0.44 AED per kWh at the margin in Dubai [17], a data-centre tariff coming in Abu Dhabi [18] and a federal rating system in drafting [19].
2. **Clean air is a warranty condition.** OEM server specifications require ISA G1 [7].
3. **The heat is already paid for.** The operator pays to reject it today.
4. **Water is scarce.** A heat-driven dryer uses no evaporative water [20, 21].
5. **Few buyers, large orders.** One procurement decision at an operator like Khazna covers hundreds of megawatts [49].

9 Market risk assumptions

These are assumptions about the market, not the build.

Assumption	Evidence for	Against or missing	Tested by
1. Operators see electrical-room humidity and corrosion as a cost worth paying to cut	ASHRAE limits, OEM G1 warranties, UAE vendors already selling filtration	No Gulf incident or spend figure found	Interviews, coupon survey
2. Operators will let a unit take heat from the GPU loop	LRZ ran heat-driven chillers nine years. Heat exchanger is isolated and bypassable	Uptime risk aversion	Interviews
3. The saving per megawatt pays back	Dubai tariff, PUE gap, a 3.4 to 5.1 year payback at Chinese prices [38]	No Gulf measurement	Coupon and RH data, vendor quote
4. Enough air-side load remains per megawatt	Air-cooled supplies, networking and storage in each rack	Unmeasured, sets market size	Interviews
5. The buyers are reachable	A few operators decide, Khazna runs its own operations	Access for students is untested	Response rate
6. Regulation will reward it	Rating system and 40 % target in drafting	Rules unpublished	Desk follow-up

The riskiest are 1 and 4: whether the problem costs operators enough, and whether enough air load remains. The MVP tests both.

10 MVP: the smallest test of demand

1. **Operator interviews, this week.** A one-page brief and three questions to the facility teams of the operators named above (Khazna, Moro Hub, Equinix Dubai, du, the Microsoft capacity through Khazna): how they control humidity and corrosion in electrical rooms today, what it costs, and how large those rooms and their airflows are per megawatt. Count replies and calls. Interviews ask about past spending and incidents, not opinions about our idea [54].
2. **Free 30-day air survey of one electrical room.** Copper and silver corrosion coupons (the ISA-71.04 method behind G1) plus a humidity logger, offered free. An operator saying yes is the commitment signal, and the result tells us whether the room meets G1 and how often it passes 60 % humidity. Coupon cost to verify.
3. **One vendor quote.** A regeneration-performance quote at 55 to 65 °C for a MOF or low-temperature silica wheel from a desiccant supplier active in the UAE [53].

Success thresholds are set by the team before the first message goes out. With a small number of operators, zero yeses out of n still allows a true rate up to $3/n$, so a single yes is informative and a few noes are not a verdict [55]. This follows the hypothesis-testing approach that a field trial of 116 startups linked to better performance and earlier pivots [56].

11 MoSCoW

- **Must.** Dried, filtered, positive-pressure air to electrical rooms meeting relative humidity below 60 %, dew point below 15 °C and ISA G1. Heat taken through an isolated, bypassable heat exchanger so IT cooling never depends on the unit. Non-chloride desiccant. Cooling coil first, desiccant as the finishing stage. Corrosion coupons and humidity sensors.
- **Should.** A dashboard reporting G1 class, humidity hours and electricity saved. A seasonal mode for winter, when loop heat runs cooler. A sizing tool per campus.
- **Could.** Extend to the air-cooled zone of the AI halls. Recover water from the regeneration exhaust, which links to our Water Hackathon work. Supply heat to adjacent offices.
- **Won't, for now.** Electricity from GPU heat (2 to 4 % [28]). Liquid chloride desiccants. Any change to the GPU loop's temperatures. Drying the whole data hall.

12 Pivot rules, set before the data

- Electrical rooms already meet G1 and stay dry cheaply: the pitch moves from protection to energy (replacing overcool-and-reheat dehumidification).
- Operators refuse access to the GPU loop: use heat already outside the IT loop, such as chiller condenser heat, with a MOF wheel suited to lower temperatures.
- Air-side load per megawatt turns out small: carry the same kit to the electrical rooms of other Gulf sites with heat and corrosive air, the strongest being sour-gas plants such as Shah, where feed gas is about 25 % H₂S [57].

13 How we got here

Five candidates were screened today against the same tests: people already spend money or time on the pain, the product delivers the outcome itself, no prior art, a sustainability link, and a demand test that can run this week.

Candidate	Finding	Status
CO2 and heat logger on school buses	Parents complain of heat, broken AC and long rides, sealed buses reach 2,500 to 5,000 ppm CO ₂ [58]. Cabin comfort ranks last by evidence of what parents already pay for, and a report reaches a parent who cannot act on it	Set aside
Pooled rides for children under 12	Ride apps exclude under-8s or unaccompanied under-12s, paid carpooling is illegal	Set aside
Gas turbine inlets	Fouling costs 10 % of power [59], coastal salt 5 to 50 times the maker's limit [60]. Exhaust-regenerated inlet desiccant already published [61]	Prior art
Solar plants	Dubai soiling 0.366 % a day [62]. Panel heat already drives sorption cooling [104]	Prior art
Substations	Transformer oil 65 to 80 °C in summer regenerating a desiccant for the substation's own air is patented [44]	Prior art
Data-centre electrical rooms	This report	Chosen

14 Limits, counterpoints and open checks

Validation status. Everything in this report is published or reported evidence. No operator has been contacted yet. The MVP in Section 10 is the first primary test.

Counterpoints a reviewer will raise.

- ASHRAE's allowable range is wide: 20 to 80 % relative humidity for class A1 equipment, up to 85 and 90 % for A3 and A4. The 60 % figure is the recommended limit and the limit for rooms with corrosive dust [2]. Operators who meet the allowable range may see no problem to solve.
- Khazna reports an annualised PUE of 1.5 or below with adiabatic free cooling, self-reported on an undated page [69]. The regional PUE gap does not hold for every operator.

- A liquid loop at W45 can reject its heat to dry coolers whenever ambient air is at or below 40 °C, about 8,344 hours a year in Abu Dhabi, and a 40 °C loop for about 6,959 hours (our count with a 5 K approach on the same weather files [14]). The heat is not stranded, it is dumped, so the proposal competes with a free alternative for that heat.
- Two AWS sites in the UAE were hit by drone strikes in March 2026, and physical-security spending may compete for the same budgets [70].
- The most-cited cost figures for contamination failures come from vendors. Roxtec states that 85 of 100 disruptive switchgear failures are partial-discharge related and that outages cost up to USD 650,000 a day [71]. These are claims, not measurements.

Open checks.

- No Gulf data-centre contamination incident, ISA-71.04 survey result or filtration spend was found. The MVP survey is meant to produce the first.
- The UPS-room humidity need is an assumption. Coolant temperatures for GB200 come from an analyst report and NVIDIA's next-generation blog, not a GB200 datasheet.
- Google Patents was unavailable for part of the search, so a formal patent search is still owed.
- Vendor figures (Uravu, Fahrenheit, Roxtec) are claims, not independent measurements.

A Industry scan detail

Industries harmed by airborne pollutants that also release energy, checked on 5 October 2026. Figures as reported in each source.

Industry	Harm from the air	Energy released	Prior art for self-protection
Gas turbines	Fouling over four months of engine data: 2.5 % efficiency, 10 % power [59]. Each 1 mbar of inlet filter pressure drop cost about 1.67 MW [63]. Coastal salt 0.05 to 0.5 ppm against makers' limit below 0.01 ppm [60]	Exhaust, high grade. A Persian Gulf gas plant study recovers 150 MW [64]	Exhaust-regenerated inlet desiccant, inlet 11.5 °C cooler [61]. Exhaust-driven absorption chilling of inlet air, 196 GWh a year [64]
Solar PV	Dubai soiling 0.366 % a day on fixed tilt, 0.156 % on trackers [62]	75 to 96 % of absorbed sunlight becomes heat [104]	Panel heat drives sorption water harvesting that cools the panel, 13 to 19 % more power [104]. Waterless robots at Noor Abu Dhabi
Substations	Salt and dust wetted by fog form conductive layers [10]. Indoor switchgear condensation causes corrosion and partial discharge [8]	Transformer losses. A 50 MVA unit loses about 152 kW at 80 % load (our arithmetic on a published datasheet). Top oil about 78 °C at 49 °C ambient in western Abu Dhabi [11]	Transformer oil regenerating a desiccant for the substation's own air [44]
Sour-gas plants	Shah feed gas about 23 to 25 % H ₂ S [57]. Sulfur gases attack copper and silver in electronics [3]	Process and utility heat	Not found. Control rooms use consumable chemical media, and regenerating an H ₂ S sorbent needs about 500 °C
Battery storage, cleanrooms	Dust on filters and coils, no Gulf failure data	Low-grade heat	Out of scope

B Earlier research rounds, key facts

Round 1, the air inside UAE school buses. With windows closed, about half of the fine particles in two Seattle school buses came from the bus's own engine, 13.4 of 26 µg m⁻³, mostly its crankcase [72]. Closed school buses

get 0.13 air changes an hour parked and 1.85 moving with ventilation off [73]. Tour buses on recirculating AC passed 3,000 ppm CO₂ [58]. Stale air from occupants at 3,000 ppm raised headache, fatigue and sleepiness in a controlled trial, while pure CO₂ at the same level did not [74]. UAE parents complain about heat, broken AC, long rides and “the smell of the bus”, and no report of fumes was found [75]. Dubai requires school bus AC to hold 24 °C at most [76]. Abu Dhabi has over 10,000 school buses and 279,000 students in 2026-27 [77]. Set aside because cabin comfort ranks last among what parents already pay for, and a report reaches a parent who cannot act on it.

Round 2, the school run. In Dubai about 44 % of students go by school bus and 47 % by private car [78]. Ride services exclude young children: Uber School and Bolt School start at 8, Careem needs an adult with children under 12, and the RTA pooled rides take 12 and over at AED 1,500 a month [79, 80]. Carpooling for a fee is prohibited in the UAE [81]. Parent-paid kid-ride startups such as Shuddle closed, while Zum and HopSkipDrive survived by switching to district-paid contracts [82]. Set aside on the team’s choice of the data-centre direction.

Methods that shaped the screening. Posted restaurant grade cards moved revenue toward A-grade restaurants and cut food-related hospitalisations 20 % [83]. Households shown their own indoor air cut it 17 %, mostly by ventilating [84], while households shown dirty indoor air in Delhi did not take up subsidised purifiers [85]. Stated willingness to pay overstates actual payment by a median factor of 1.35 in one meta-analysis [86] and by about 3 in another [87], which is why the MVP counts commitments, not opinions.

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